

Module -1

DC Circuit Analysis and Theorems

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SENSE

Superposition Theorem

The idea of superposition depends on the linearity property.

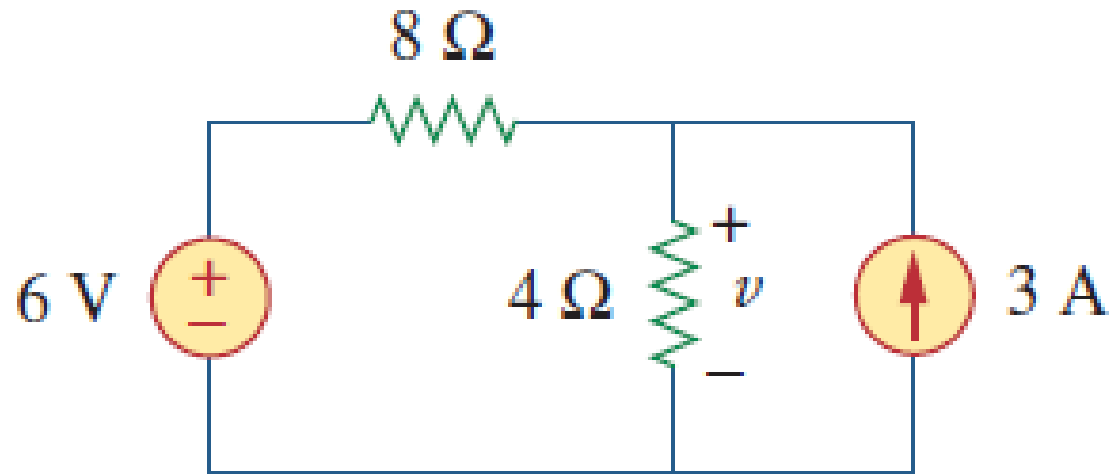
The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

Steps:

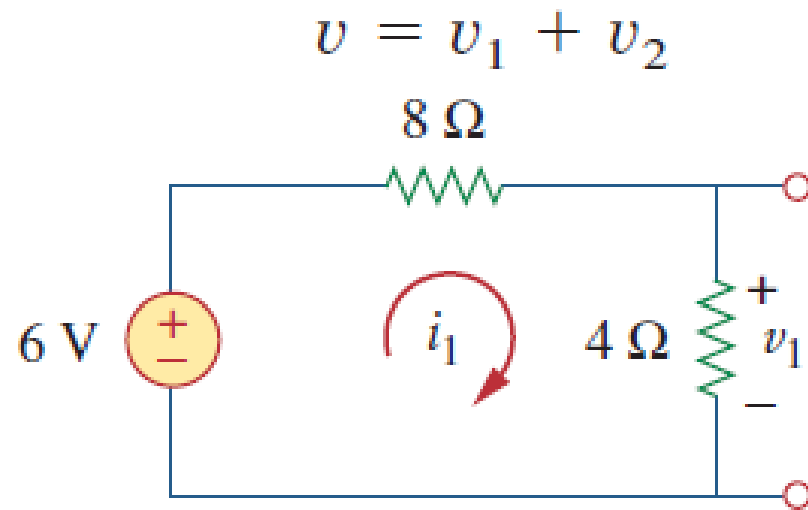
1. Turn off all the independent sources except one source. Find the output (voltage or current) due to that active source using any techniques.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

Tutorial - 1

Use the superposition theorem to find v in the circuit.



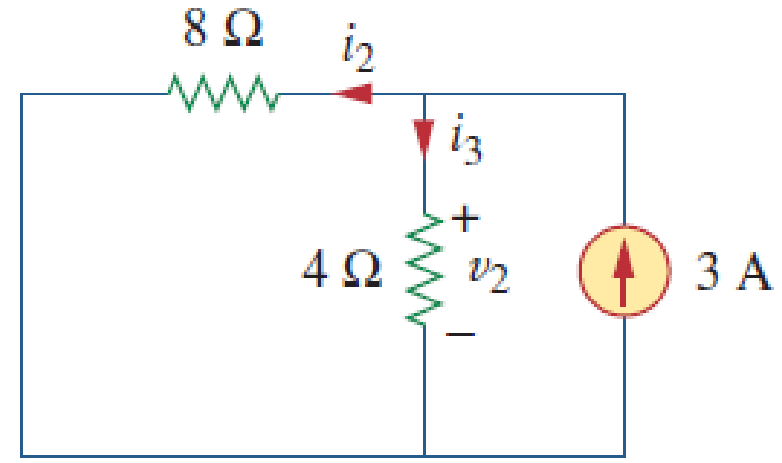
Solution:



$$12i_1 - 6 = 0 \quad \Rightarrow \quad i_1 = 0.5\text{ A}$$

$$v_1 = 4i_1 = 2\text{ V}$$

$$v_1 = \frac{4}{4 + 8}(6) = 2\text{ V}$$



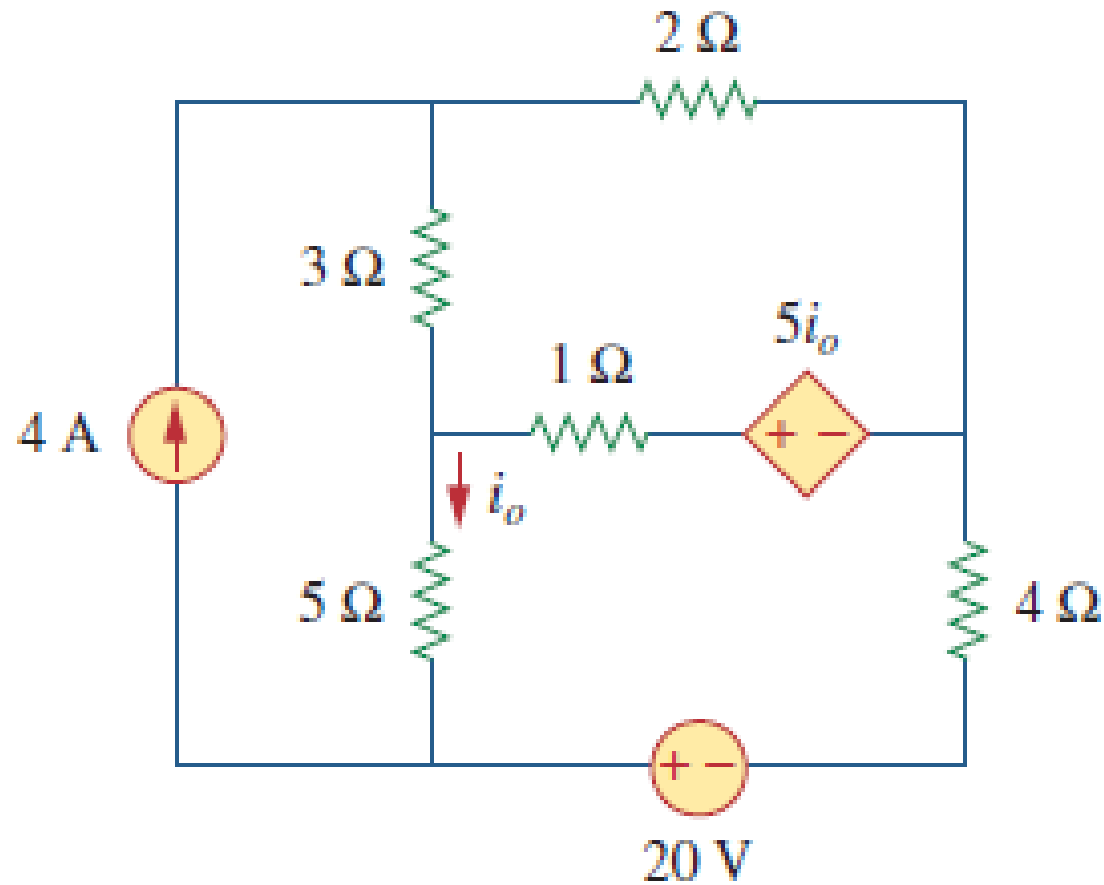
$$i_3 = \frac{8}{4 + 8}(3) = 2\text{ A}$$

$$v_2 = 4i_3 = 8\text{ V}$$

$$v = v_1 + v_2 = 2 + 8 = 10\text{ V}$$

Tutorial - 2

Determine I_o in the circuit using superposition theorem.



Solution:

$$i_o = i'_o + i''_o$$

For loop 1,

$$i_1 = 4 \text{ A}$$

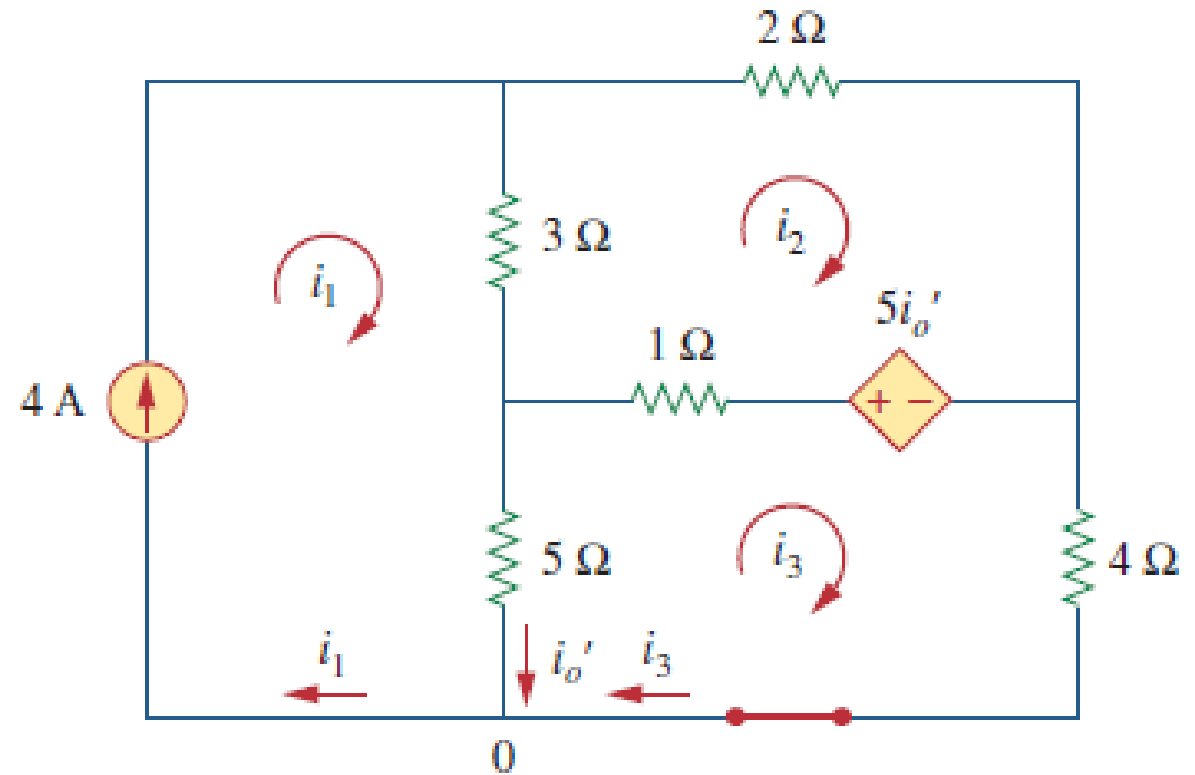
For loop 2,

$$-3i_1 + 6i_2 - 1i_3 - 5i'_o = 0$$

For loop 3,

$$-5i_1 - 1i_2 + 10i_3 + 5i'_o = 0$$

But at node 0, $i_3 = i_1 - i'_o = 4 - i'_o$



$$\begin{aligned} 3i_2 - 2i'_o &= 8 \\ i_2 + 5i'_o &= 20 \end{aligned} \quad i'_o = \frac{52}{17} \text{ A}$$

For loop 4,

$$6i_4 - i_5 - 5i''_o = 0$$

for loop 5,

$$-i_4 + 10i_5 - 20 + 5i''_o = 0$$

But $i_5 = -i''_o$. $6i_4 - 4i''_o = 0$

$$i_4 + 5i''_o = -20$$

$$i''_o = -\frac{60}{17} \text{ A}$$

$$i_o = -\frac{8}{17} = -0.4706 \text{ A}$$

